

# DNA CELL CYCLE STUDIES IN UVEAL MELANOMA\*

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## INTRODUCTION

UVEAL MELANOMAS HAVE BEEN TREATED WITH A NUMBER OF RADIATION modalities, including pre-enucleation photon irradiation, brachytherapy with different radioactive isotopes, and either helium ion or proton charged-particle beams.<sup>1-22</sup> The goal of all forms of radiation therapy is to destroy the reproductive integrity of a tumor; however, it is often difficult to determine whether this objective has been achieved.

Clinical, fluorescein angiographic, and ultrasonographic data often do not accurately predict or document radiation tumor control. After successful radiation of uveal melanomas with either cobalt plaques or charged-particle beams, the mean tumor shrinkage is approximately 40%; only 10% of tumors are reduced to a flat scar.<sup>5,16,18,21,22</sup> Occasionally, uveal melanoma necrosis has been reported after treatment despite little or no clinical or ultrasonographic evidence of radiation response.<sup>23</sup>

In a number of human and animal malignancies, apparently viable tumor cells remain after radiation sterilization of the neoplasm.<sup>24</sup> Most uveal melanomas studied after brachytherapy or charged-particle irradiation have little or no histologic evidence of mitotic activity. Nonetheless, they often contain viable-appearing tumor cells.<sup>25-32</sup> In studies of several human malignancies, histologically monitored mitotic activity did not correlate with tumor growth.<sup>33,34</sup> In tumors such as uveal melanoma that have relatively slow doubling times, histologic studies of mitotic activity are insensitive measures of cell cycling.<sup>35-39</sup>

A number of constraints have restricted the usefulness of tissue culture techniques to study the effectiveness of radiation therapy in uveal melanoma. Most laboratories have not been able to establish long-term continuous tissue culture lines from uveal melanomas.<sup>40-46</sup> The majority of explants

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grow for a very short time, require either continuous feeder layers or artificial substrates, and cannot be manipulated in a manner analogous to other tumor cell lines.<sup>47-49</sup> These approaches have other limitations. The tissue culture cell line may be established from a nonrepresentative clone of cells, and fibroblasts often overgrow the culture.<sup>50</sup> The radiation response of a cell line does not accurately reflect the response of fresh melanoma tissue.<sup>51</sup>

Methodologic advances in flow cytometry and monoclonal antibody tumor cell labeling may be useful to elucidate the effect of radiation on the reproductive status of uveal melanoma cells. Flow cytometry (FACS) requires cell dispersion, quantitative staining, and high-resolution photometric analysis.<sup>52</sup> A number of fluorochromes can be used to label total deoxyribonucleic acid (DNA) and delineate stages of cell reproduction.<sup>53,54</sup>

The cell cycle can be divided into four sequential phases: mitosis (M), gap 1 (G1), DNA synthesis (S), and gap 2 (G2).<sup>55</sup> Since the intensity of fluorescent emission is proportional to DNA content, it can be used to estimate the percentage of cycling, noncycling, and dead cells.<sup>52-56</sup> However, statistical methods based on these techniques overestimate G1 phases and underestimate S phases. It has also been impossible to differentiate early S cells from G1 cells or late S phase cells from G2/M phase cells.<sup>57-59</sup>

The introduction of a thymidine analog, bromodeoxyuridine (BrdUrd), incorporated only during the S phase of the cell cycle, and the use of anti-BrdUrd monoclonal antibody have simplified the problem of differentiating cycling cells from noncycling cells.<sup>39,60-62</sup> BrdUrd competes with both intracellular thymidine and the thymidine precursor, thymidine monophosphate, for incorporation into the DNA during the S phase of the cell cycle.<sup>63,64</sup> The use of BrdUrd is an extremely sensitive method to label cells in S phase, especially in relatively slow-growing tumors.<sup>39</sup> In both experimental studies and mathematical models, the BrdUrd uptake is stoichiometrically related to the number of cells in S phase.<sup>39,65-67</sup>

A number of investigators have demonstrated that measurement of S phase by BrdUrd incorporation correlates quite closely with assessment using either autoradiography or <sup>3</sup>H-thymidine assays.<sup>68-70</sup> Unlike the latter two assays, BrdUrd studies can be done rapidly, are better for delineation of S phase, and have no radiation toxicity for *in vivo* clinical studies.

Treatment affects the cell cycle. DNA cell cycle analysis may improve our understanding of the effects of radiation on uveal melanoma. If radiation therapy is completely effective, the number of cells in S phase should eventually be reduced to near zero and cells should stop cycling.<sup>24</sup> Cell cycle studies may delineate the reproductive integrity of these cells and the effect of radiation on their ability to cycle. In this investigation, we have

used in vivo BrdUrd techniques to study the effects of two types of radiation on the cell cycle of uveal melanoma cells. We have observed a significant reduction in DNA S after either 20 gray (Gy) of pre-enucleation or > 60 GyE of helium ion charged-particle irradiation.

## METHODS

### PATIENT SELECTION

This study involved a group of patients treated between 1983 and 1986. All patients had histologically documented choroidal or ciliary body melanomas. After the protocol was approved by the human experimentation committee and written and verbal informed consent was obtained, individuals were enrolled in the study. Tumors were removed (by enucleation or cilio-choroidectomy) after intravenous injection of BrdUrd 200 mg/m<sup>2</sup> body surface, 1 hour prior to surgery. Complete follow-up data were available, and adequate fresh and paraffin-embedded material could be analyzed in each case. Twenty-three patients met the criteria and were included in the study. One patient had bilateral primary uveal melanomas, and the results of both eyes are reported.

Table I summarizes the clinical, ultrasound, and irradiation parameters for patients in the study. The control group consisted of melanomas treated with surgery only. The experimental groups consisted of melanomas treated with either photon or helium ion radiation. For patients receiving helium ion radiation, the radiation dose, interval between radiation and enucleation, and rationale for enucleation are also described in Table I. Nine eyes were studied after primary enucleation, and four tumor specimens were obtained by ciliochoroidectomy. Five eyes were treated for 5 days prior to enucleation with 20 Gy of photon irradiation (five daily fractions of 400 centigray [cGy], usually followed by enucleation within 48 hours). Six eyes were studied after enucleation following helium ion therapy. Enucleation after helium ion therapy was always due to complications of radiation, not to evidence of tumor growth. The mean interval between charged-particle irradiation and enucleation was 25 months, with a range of 13 to 48 months (Table I).

### TISSUE PROCESSING

Most of the melanoma tissue was fixed in 10% formalin and processed in the conventional manner for standard light microscopic analysis. Small pieces were obtained for tissue culture propagation, and fresh tissue studies were obtained from 15 patients. In these latter cases, dispersed cells were fixed

TABLE I: UVEAL MELANOMA PATIENT SUMMARY

| PATIENT | AGE | TUMOR<br>DIAMETER | ULTRASOUND HEIGHT |       | RADIATION   | DATE OF RADIA-<br>TION COMPLETION | DOSE (Gy) | DATE OF<br>ENUCLEATION |
|---------|-----|-------------------|-------------------|-------|-------------|-----------------------------------|-----------|------------------------|
|         |     |                   | INITIAL           | FINAL |             |                                   |           |                        |
| MM      | 63  | 13.00             | 12.26             |       | None        |                                   |           | 08/12/85               |
| HS      | 58  | 14.00             | 7.09              |       | None        |                                   |           | 12/02/85               |
| BR      | 66  | 15.00             | 10.72             |       | None        |                                   |           | 02/10/86*              |
| CW      | 79  | 12.00             | 10.34             |       | None        |                                   |           | 02/13/86               |
| JB      | 78  | 14.60             | 7.09              |       | None        |                                   |           | 04/14/86*              |
| AP      | 79  | 12.00             | 4.40              |       | None        |                                   |           | 06/05/86*              |
|         | 79  | 10.50             | 7.09              |       | None        |                                   |           | 11/03/86*              |
| MH      | 22  | 18.00             | 13.41             |       | None        |                                   |           | 06/16/86               |
| AS      | 73  | 12.00             | 3.45              |       | None        |                                   |           | 07/21/86               |
| DD      | 71  | 18.00             | 9.96              |       | None        |                                   |           | 08/06/86               |
| DB      | 42  | 17.00             | 13.50             |       | None        |                                   |           | 08/21/86               |
| WB      | 61  | 12.00             | 7.66              |       | None        |                                   |           | 11/17/86               |
| BL      | 66  | 15.00             | 11.87             |       | None        |                                   |           | 03/17/85               |
| JH      | 67  | 22.50             | 7.47              |       | Photon†     | 08/12/85                          | 20        | 08/13/85               |
| BW      | 67  | 15.00             | 9.19              |       | Photon†     | 03/11/86                          | 20        | 03/13/86               |
| RG      | 69  | 20.00             | 12.45             |       | Photon†     | 05/23/86                          | 20        | 05/29/86               |
| IA      | 56  | 18.00             | 14.50             |       | Photon†     | 06/06/86                          | 20        | 06/09/86               |
| EG      | 75  | 18.00             | 9.96              |       | Photon†     | 08/12/86                          | 20        | 08/14/86               |
| CF      | 64  | 18.00             | 8.62              | 8.62  | Helium ion‡ | 11/16/84                          | 70        | 12/02/85               |
| NO      | 76  | 24.00             | 8.23              | 8.23  | Helium ion‡ | 10/31/84                          | 70        | 03/13/86               |
| CM      | 52  | 13.00             | 6.32              | 4.79  | Helium ion‡ | 07/02/84                          | 70        | 08/18/86               |
| JW      | 53  | 13.00             | 9.50              | 3.26  | Helium ion‡ | 02/12/85                          | 70        | 07/18/86               |
| LF      | 71  | 7.00              | 5.17              | 3.64  | Helium ion‡ | 09/07/82                          | 80        | 09/24/86               |
| ES      | 69  | 10.50             | 8.23              | 5.17  | Helium ion‡ | 02/09/84                          | 60        | 10/22/86               |

\*Ciliochoroidectomy.

†Treated for 5 days prior to enucleation.

‡Rationale for enucleation for patients receiving helium ion radiation: CF, intractable pain, neovascular glaucoma; NO, intractable pain, no light perception; CM, intractable pain, neovascular glaucoma; JW, intractable pain, no light perception; LF, intractable pain, neovascular glaucoma, no light perception; ES, intractable pain, descemetocoele.

in ethanol prior to analysis of cell cycling (see below).

Paraffin-embedded 8  $\mu$  sections were cut, placed on glass slides, and stained for BrdUrd uptake as follows. The paraffin was removed from the sections with two 10-minute washes in xylene. The sections were then rehydrated through 10-minute washes in 100% EtOH, 95% EtOH, 70% EtOH, and distilled water. The DNA was denatured by incubating slides for 20 minutes in 2 M HCl and 0.5% triton at room temperature, then washing the slides twice in phosphate-buffered saline (PBS). The slides were then treated with 0.5% pepsin in 0.9 M NaCl, pH 1.5, for 30 minutes at room temperature. The slides were washed twice with PBS, then incubated with two different mouse anti-BrdUrd antibodies (Lawrence-Livermore Laboratories) in a mixture of PBS 1%, bovine serum albumin, and 0.5% Tween (PBS/BSA/Tween) for 30 minutes at room temperature.<sup>71,72</sup> The slides were washed twice with this solution (PBS/BSA/Tween) and incubated in the dark with fluorescein (FITC)-tagged goat anti-mouse IgG for 30 minutes. Slides were then washed twice in PBS/BSA/Tween, cover slipped, and observed on a Zeiss fluorescence microscope at 40 $\times$  under oil. A positive-control retinoblastoma cell line, labeled in vitro with BrdUrd, was used to determine uniformity of BrdUrd staining with anti-BrdUrd monoclonal antibody. Under proper conditions, approximately 30% of retinoblastoma tissue culture cells on each slide stained with BrdUrd.

All slides were read in a masked manner. Propidium iodide (PI) counterstain (red) was used to enhance contrast with the fluorescent (green) antibody stain. Light microscopy of the PI-stained tissues was used to choose areas of viable-appearing cells in which to study BrdUrd uptake. Cells with disrupted cellular membranes, absent nuclei, or other evidence of necrosis were not counted. Prior to enumeration of BrdUrd-positive cells using the immunofluorescence, we eliminated areas of extreme necrosis, because they would not have viable, cycling cells. Because tumors have areas with variable labeling indices, we attempted to decrease false-negative data with this approach. A PI-stained slide is shown in Fig 1; this stain gave adequate resolution to detect areas of necrosis and did not interfere with the immunofluorescence of the anti-BrdUrd antibody. Fig 2 shows a slide that stained positively with anti-BrdUrd antibody. BrdUrd labeling was expressed as the total number of positively stained, viable-appearing tumor cells in 32 random 40 $\times$  fields.

A 1-mm<sup>3</sup> portion of fresh melanoma tumor tissue was mechanically dispersed by mincing and then vigorously pipetting in PBS. Cell viability was immediately checked with the trypan blue exclusion assay. Dead cells have permeable cell membranes and the cell stains blue, while viable cells

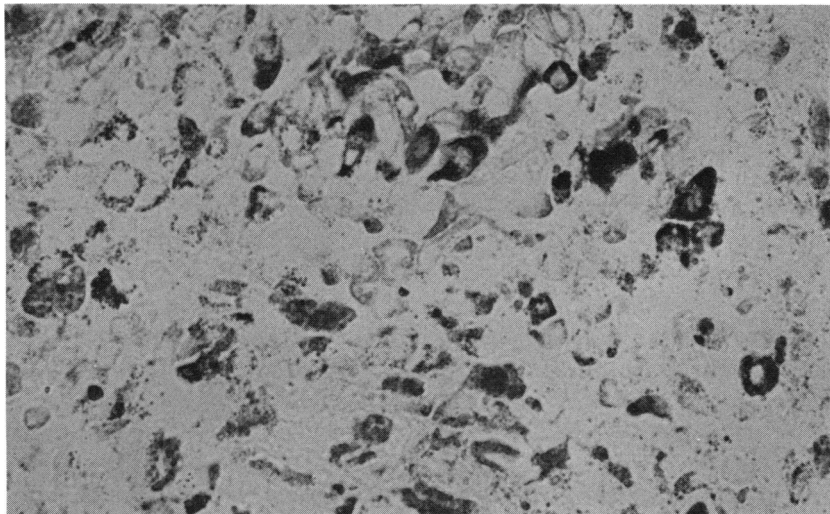


FIGURE 1

Visible light microscopy of tissue stained with PI counterstain. Morphologic detail is visible. Prior to immunofluorescent study, all melanomas were screened with light microscopy to aid in choosing fields that had the maximum number of viable cells. This approach was selected to minimize false-negative BrdUrd uptake, since necrotic cells do not cycle.

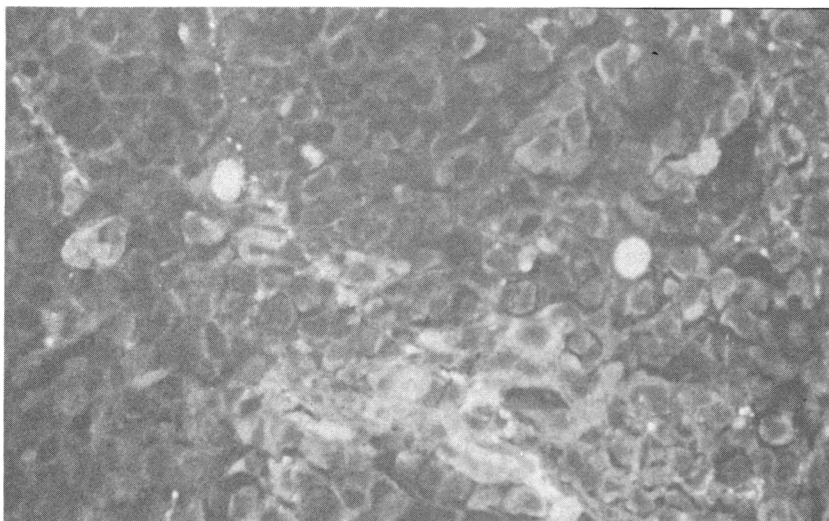


FIGURE 2

Fluorescent microscopy showing a melanoma staining positive with BrdUrd. Three positive staining cells are seen in this high-power field. Positive cells are in the DNA S phase (40 $\times$  magnification).

exclude this stain. Differentiation of these two populations is easily visible on standard light microscopic examination. Cells were pelleted and then resuspended in 70% EtOH and fixed for 12 hours. Cells were then rehydrated with washes of 50% EtOH and distilled water. DNA denaturation and BrdUrd staining were performed as previously described. Cells were washed twice, spotted on slides, and observed with a Zeiss fluorescence microscope. This approach was used as another control to determine BrdUrd labeling in cases in which the eye was opened prior to fixation. This alteration in fixation technique was incorporated in these studies to assess whether the type of fixation used had an adverse effect on BrdUrd analysis. No significant differences were noted.

In ten cases, we obtained fine-needle biopsy specimens with a 25-gauge needle. The cells were placed in PBS, and their viability was assessed with trypan blue exclusion. Cells were then centrifuged and the pellet vortexed and fixed in 70% EtOH. These were processed for BrdUrd studies in the same way as fresh tissue (described previously).

In 15 cases, a small piece of fresh melanoma tissue was obtained in a sterile manner and transported to our laboratory in a tube containing 10 ml of RPMI 1640 medium supplemented with 15% fetal calf serum, glutamine, penicillin, and streptomycin. The tissue was dispersed into a suspension of single cells and small clumps by vigorous pipetting. Cell viability was assessed using the trypan blue exclusion assay. The percentage of viable cells in the original tumor was defined as those cells that excluded trypan blue (Table II). The cells were then placed in 25-cm<sup>2</sup> tissue culture flasks in 15 ml of the RPMI 1640 medium described above and incubated at 37°C with 5% CO<sub>2</sub>. Flasks were fed once a week by removing 5 ml of spent medium and replacing it with fresh medium. When cultures reached confluency, they were passaged by trypsin treatment, diluted to one-third confluency, and seeded into three 25-cm<sup>2</sup> flasks. Cultures were followed until growth stopped or fibroblasts overtook the culture. Table II shows the results of tissue culture studies.

Cells analyzed by flow cytometry had two additional preparatory steps. Immediately preceding DNA denaturation, cells were incubated in 1 ml of RNAase A (Sigma Chemical Co, St Louis, MO), 1 mg/ml in PBS at 37°C for 30 minutes, then washed once in 10 ml of PBS. After the final PBS/BSA/Tween wash, cells were counterstained in 1 ml of PI (10 µg/ml in PBS) for at least 30 minutes before dual-parameter analysis on the flow cytometer.

#### FLOW CYTOMETRY

Samples studied by flow cytometry were run on a FACS II flow cytometer in list mode. The data were analyzed on a VAX 11/750 minicomputer.

TABLE II: TISSUE CULTURE DATA ON UVEAL MELANOMA EXPLANTS\*

| PATIENT | NO. OF PASSAGES | TIME IN CULTURE (WK) | TREATMENT               | ORIGINAL TUMOR VIABILITY (%) <sup>†</sup> |
|---------|-----------------|----------------------|-------------------------|-------------------------------------------|
| MM      | 12              | 20                   | Enucleation only        | 87                                        |
| HS      | 2               | 5                    | Enucleation only        | 39                                        |
| BR      | 5               | 13                   | Enucleation only        | 40                                        |
| JB      | 15              | 14                   | Enucleation only        | 60                                        |
| MH      | 15              | 14                   | Enucleation only        | 32                                        |
| AS      | 3               | 8                    | Enucleation only        | 46                                        |
| BL      | 5               | 14                   | Enucleation only        | 39                                        |
| JH      | 1‡              | 4                    | Preoperative radiation§ | 47                                        |
| BW      | 1‡              | 4                    | Preoperative radiation§ | 3                                         |
| RG      | 1‡              | 4                    | Preoperative radiation§ | 84                                        |
| IA      | 7               | 14                   | Preoperative radiation§ | 36                                        |
| EG      | 3               | 6                    | Preoperative radiation§ | 11                                        |
| CF      | —               | —                    | Helium ion¶             | 0                                         |
| NO      | —               | —                    | Helium ion¶             | 0                                         |
| CM      | —               | —                    | Helium ion¶             | 50                                        |

\*Growth rates of melanoma cells in culture were not constant but generally tended to decline steadily during the culture period, and often the cultures were overtaken by fibroblasts. Two explants, MM and MH, yielded stable cultures with reasonable growth rates, and the remainder were discontinued and stored in liquid nitrogen.

†Determined immediately after surgery by trypan blue exclusion (see methods).

‡Cells did not reach confluence on passaging.

§20 Gy preoperative radiation.

||Cells never grew to confluency.

¶>60Gyhelium ion radiation.

Photomultiplier tubes detected fluorescein and PI fluorescence and forward scatter. The laser was set at 488 nm to excite fluorescein; emissions were detected through a  $530 \pm 30$  nm filter. PI was excited at 488 nm, and emissions were detected through a 580-nm long pass filter. Ten to 20 thousand cells were counted. Statistical analysis was performed on a CRISP computer program using Fisher's exact test, Kendall's Tau B, or Pearson's correlation coefficient.

## RESULTS

As shown in Table III, there were more melanoma cells in S phase in untreated (control) than in irradiated (treated) uveal melanomas ( $P < .0001$ ). The mean BrdUrd count in 32 high-power fields in 13 melanomas removed without radiation was 79.5. In contrast, the mean number of cells staining



TABLE III: NO. OF BrdUrd-POSITIVE CELLS  
IN UVEAL MELANOMAS IN 32 RANDOM  
HIGH-POWER FIELDS

| PATIENT                                                                | NO. |
|------------------------------------------------------------------------|-----|
| Uveal melanomas receiving<br>no radiation                              |     |
| MM                                                                     | 4   |
| HS                                                                     | 27  |
| BR                                                                     | 89  |
| CW                                                                     | 69  |
| JB                                                                     | 54  |
| AP                                                                     | 20  |
| MH                                                                     | 329 |
| AS                                                                     | 86  |
| DD                                                                     | 298 |
| DB                                                                     | 123 |
| JB                                                                     | 7   |
| BL                                                                     | 2   |
| Uveal melanomas treated<br>with 20 Gy of pre-<br>enucleation radiation |     |
| JH                                                                     | 0   |
| BW                                                                     | 0   |
| RG                                                                     | 0   |
| IA                                                                     | 0   |
| EG                                                                     | 4   |
| Uveal melanomas treated<br>with helium ion<br>radiation*               |     |
| CF                                                                     | 0   |
| NO                                                                     | 0   |
| CM                                                                     | 0   |
| JW                                                                     | 0   |
| LF                                                                     | 0   |
| ES                                                                     | 0   |

\*All patients had tumor regression; enuclea-  
tion was required because of radiation com-  
plications, not tumor growth.

with BrdUrd after 20 Gy of pre-enucleation radiation was 0.8 in the five tumors studied. In six eyes enucleated after helium ion irradiation, without evidence of posttreatment tumor growth, the mean count was 0.

In nine melanomas we compared the BrdUrd values obtained with more than one technique. Fluorescence microscopy on both whole sections and fine-needle aspirates of the same tumor was compared with the BrdUrd results using two-directional flow cytometry. Generally, the results in each tumor were similar (Table IV). Tumors with relatively low numbers of cells in S phase had low counts with both methods, while tumors with high num-

TABLE IV: COMPARISON OF BrdUrd UPTAKE VALUES OBTAINED WITH DIFFERENT TECHNIQUES

| PARAFFIN-EMBEDDED<br>WHOLE SECTIONS |                                                                        | FRESH TISSUE FIXED IN ETHANOL  |                            |                                   |               |
|-------------------------------------|------------------------------------------------------------------------|--------------------------------|----------------------------|-----------------------------------|---------------|
|                                     | NO. OF BrdUrd-<br>POSITIVE CELLS IN<br>32 RANDOM HIGH-<br>POWER FIELDS | % POSITIVE FOR BrdUrd STAINING |                            |                                   |               |
|                                     |                                                                        | FLOW CYTOMETRY                 | FLUORESCENCE<br>MICROSCOPY | FLUORESCENCE<br>MICROSCOPY (FNB)* |               |
| JB                                  | 54                                                                     | 1.09 (whole)<br>1.62 (FNB)     | 1.1                        | 1.1                               | 1.14 (2 FNBs) |
| MH                                  | 329                                                                    |                                | 0.61                       | 0.92                              |               |
| AS                                  | 86                                                                     |                                | 2.1                        | 2.0                               |               |
| BL                                  | 2                                                                      |                                |                            | 0                                 |               |
| BW                                  | 0                                                                      |                                | 0                          | 0                                 |               |
| EG                                  | 4                                                                      |                                | 0.01                       | 0                                 |               |
| RG                                  | 0                                                                      |                                | 0                          | 0.4                               |               |
| NO                                  | 0                                                                      |                                |                            | 0                                 |               |
| CM                                  | 0                                                                      |                                | 0.01                       | 0                                 |               |

\*FNB, fine-needle biopsy.

bers of cells in S phase had higher BrdUrd uptake with both techniques. We chose to rely on our microscopy data, since they allowed us to eliminate artifacts caused by either staining of debris or nontumor cells infiltrating the melanoma.<sup>73</sup>

Nonparametric analysis was used to study possible correlations between cell type, tumor diameter, quantitative ultrasonographic thickness, histologic evidence of M per high-power field, length of time between radiation and enucleation, and BrdUrd uptake. We observed no correlation between any of these factors and BrdUrd uptake.

We compared the fine-needle biopsy specimen BrdUrd results with standard histologic sections and flow cytometry of the same tumor. Results were similar (Table IV), although the sample size is too small to draw significant conclusions.

We were unable to propagate any long-term, rapidly growing tissue culture cell lines from the uveal melanoma specimens. In seven of seven non-irradiated cases, the uveal melanoma explants remained viable for 3 to 15 passages. Two of five samples from tumors that received 20 Gy of pre-enucleation radiation were maintained for three and seven passages; the other three tumors failed to grow in vitro. None (0 of 3) of the helium ion-irradiated samples could be established as a replicating uveal melanoma explant (Table II). The difference between radiated and nonirradiated tissue culture results was significant (Fisher's exact test,  $P < .007$ ).

#### DISCUSSION

Few data are available on DNA S rates in human malignancies.<sup>74-79</sup> BrdUrd

labeling, as a measure of S phase, was relatively low (1% to 8%) in nonirradiated uveal melanomas investigated in this study; these data imply a relatively slow doubling rate for this tumor. In contrast, between 20% and 35% of viable retinoblastoma cells are in S phase (unpublished observations). These results are consistent with other reports in which epithelial and endodermal malignancies appear to have a higher DNA S rate than most mesenchymal neoplasms, such as sarcomas or neuroectodermally derived melanomas.<sup>74,76</sup>

The relationship of mitotic rate to radiation response has been studied since 1906.<sup>80</sup> Radiation, at different fractionation and dose levels, can result in immediate cell death, sterilization, or proliferation delay.<sup>24</sup> In the interval between radiation and eventual cell destruction, giant cell formation, mitotic delay, and chromosomal aberrations can occur.<sup>81</sup> Radiation can inhibit entry of cells into the M phase of the cell cycle<sup>2</sup>; this cell cycle block usually occurs either at the G1/S or G2/M phase of the cycle.<sup>24,82</sup> At moderate doses of radiation (< 20 Gy), tumor cells often remain viable and may complete one or two divisions before death.<sup>24</sup> The most common lethal lesions from radiation therapy are double-stranded DNA breaks associated with chromatid exchange, acentric or dicentric chromosomes, or micronuclear fragments.<sup>83</sup> Generally, over 99% of tumor cells lose their reproductive integrity after treatment with more than 20 Gy of radiation.

It is often difficult, clinically or with standard histologic techniques, to determine whether radiation has altered the reproductive integrity of uveal melanoma cells.<sup>84</sup> A number of investigators have demonstrated that tumor cells can remain in situ after successful radiation destruction of reproductive capacity, and that histologic evidence of mitosis is an insensitive gauge of tumor cell proliferation.<sup>33,34</sup> In this study we have shown that the uveal melanoma cell cycling, as measured by cells in S phase, appears to have stopped after 20 Gy of pre-enucleation radiation or over 60 GyE of helium ion therapy. Tissue culture findings support these results. Radiated melanoma cells were less capable of forming expanding explants than nonirradiated melanomas.

These studies have a number of potential ramifications. We and others<sup>85-89</sup> have used fine-needle biopsy to diagnose a number of orbital and ocular tumors. In a few cases we have analyzed specimens from fine-needle biopsies using a modification of the BrdUrd uptake studies described in this report. While fine-needle biopsy has a number of problems, the most important being potential sampling errors, such studies could be important in at least four settings.<sup>90-92</sup>

First, we have previously demonstrated that alterations in DNA content are more closely correlated with survival than either cell type or tumor

size.<sup>93</sup> Fine-needle biopsy and assessment of DNA aneuploidy could be a useful technique to increase the accuracy of pretreatment prognosis assessment of uveal melanoma patients who are not treated with enucleation.

Second, in uveal melanomas in which growth is equivocal after irradiation, a fine-needle biopsy with BrdUrd studies may ascertain whether the tumor is still cycling or whether growth could reflect intratumor hemorrhage, edema, or an inaccurate evaluation of size change.<sup>94</sup> In two patients, we have used this approach clinically. One patient had a 12-mm-thick posterior uveal melanoma in his only eye; it had been treated with a cobalt plaque elsewhere. He was referred for management of recurrent tumor growth. Fine-needle biopsy revealed no cells in S phase on the basis of BrdUrd stains, and he has been followed without change in tumor size for 2 years. A second patient had been treated elsewhere with 60 Gy of photon radiation for a melanoma in his only eye. Possible growth was noted, and he was referred for evaluation. A fine-needle biopsy after BrdUrd infusion revealed numerous cells in S phase. The eye was enucleated because of these findings, and numerous mitoses per high-power field were demonstrated.

Third, in some malignancies, response to either radiation or chemotherapy has correlated with the number of cells in S phase and the degree of DNA aneuploidy.<sup>95,96</sup> These data might be important in treatment planning. As an example, some orbital and intraocular tumors are radioresponsive, while others are not.<sup>97-103</sup> BrdUrd studies of a fine-needle biopsy might result in an accurate prediction of radiation response and avoid unnecessary radiation exposure in patients with radioresistant lesions. Similarly, radiation dose and fractionation schedules might be altered on the basis of DNA aneuploidy and S phase data in some uveal melanomas. Conceivably, the wide-range regression patterns after uveal melanoma irradiation may partially reflect differences in these tumor parameters. Some uveal melanomas could be treated with less radiation, on the basis of their DNA content or number of cells in S phase, with less resultant ocular morbidity.

Fourth, cell doubling time can be estimated using a single injection of BrdUrd.<sup>104,105</sup> As we and others have stressed, little information is available on uveal melanoma growth patterns or doubling times.<sup>36-38</sup> Investigators have based major conclusions regarding the efficacy of different forms of uveal melanoma therapy on tenuous tumor doubling data.<sup>38,106</sup> More accurate data would be helpful in understanding growth and metastatic patterns of uveal melanoma and therefore in planning therapies. Our data, demonstrating that most nonirradiated uveal melanomas had less than 2% of cells in S phase, would suggest that they grow relatively slowly. These

data would be consistent with some of the mathematical models that have been proposed.<sup>107,108</sup> However, tumor growth is not constant. Further studies in melanomas of different size are needed, since most of our investigations were performed on relatively large neoplasms.

BrdUrd assays have a number of potential limitations. Our studies do not address all of them. First, BrdUrd assessment of S phase has some theoretic and actual limitations. BrdUrd may not be uniformly incorporated into DNA during the entire S phase of the cell cycle.<sup>109</sup> DNA S can occur outside of S phase during repair of sublethal damage as a result of either radiation or chemotherapy.<sup>109</sup> In some neoplasms, it is difficult to determine whether failure of BrdUrd incorporation is due to absence of DNA S or nonavailability of BrdUrd either as a result of poor vascular supply to the tumor or limited tissue diffusion.<sup>68</sup>

A major problem with flow cytometric analysis of S phase, and the predominant reason we chose to base our studies on the fluorescent microscopy data, is that nontumor cells infiltrating a malignancy are also stained with BrdUrd if they are cycling.<sup>73,110</sup> While inflammatory cell infiltration into uveal melanomas is relatively uncommon, it does occur. Unless dual- or three-channel flow cytometry is performed with a different fixation technique and additional labeled monoclonal antibodies directed toward antigens on lymphocytes and monocytes, microscopy is needed to screen out BrdUrd uptake by nontumor cells.<sup>111,112</sup> Similarly, it can be difficult to avoid other staining artifacts. With microscopy studies, we could be more certain we were selecting viable-appearing melanoma cells and could attempt to optimize the number of cells that stained with BrdUrd and thus avoid potential false-negative results.

The requirement for DNA denaturation to label cells with BrdUrd is also a potential pitfall of this technique. While few data prove that denaturation is nonuniform during different parts of the cell cycle, this is a possibility.<sup>111</sup> It is also theoretically possible that cell dispersal techniques, necessary for flow cytometric analysis, could preferentially select viable cells.<sup>113</sup> This latter possibility is probably an advantage, since cycling cells are the important cells to analyze.

Finally, the accuracy of BrdUrd uptake as a measure of radiation destruction of a tumor's reproductive integrity depends on appropriate sampling. At radiation doses of 20 Gy or more, over 99% of tumor cells are probably sterilized.<sup>24</sup> In studies of uveal melanoma cells performed 48 hours after receiving 20 Gy of pre-enucleation radiation, probably less than 1% of cells can potentially cycle and some of these cells may temporarily be arrested in the cell cycle.<sup>24</sup> It is difficult to determine if this small number of cells was sampled or reflected permanent versus transient damage at the

time of study. Conceivably, if the interval between 20-Gy photon irradiation and enucleation were increased to several months, the presumed 1% of cells that may have retained reproductive integrity could have reexpanded and a number of these might have stained with BrdUrd. We have clinically examined uveal melanomas treated with 30 Gy of photon irradiation that showed renewed tumor growth within 1 year of treatment, presumably on this basis. The results observed in the melanomas treated with helium ions should accurately reflect destruction of tumor reproductive integrity. These neoplasms were enucleated, on average, over 2 years after radiation without evidence of tumor reactivation. In all cases, multiple sections failed to demonstrate BrdUrd uptake.

The morbidity of *in vivo* BrdUrd studies has been assessed and appears to be minimal. At our institution, we have studied over 150 patients with various types of malignancies who had no obvious side effects with this agent.<sup>34,114</sup> The mean plasma concentration of this agent is 8  $\mu\text{mol}$  2 hours after treatment. Further, in a number of previous studies, BrdUrd was given in much higher doses for up to several weeks at a time as a radiation sensitizer. In these studies, there was no permanent morbidity.<sup>33,114,115</sup>

This approach may be very useful in increasing our understanding of the biology, as well as the management, of uveal melanomas and other intraocular and orbital tumors.

#### SUMMARY

We analyzed the cell cycling status of a group of irradiated and nonirradiated uveal melanomas using BrdUrd techniques. These data demonstrate that melanomas are relatively slow-growing tumors with a few cells actively cycling at a given time. Radiation has a profound effect on the number of cycling cells ( $P < .0001$ ). After treatment with either 20 Gy of pre-enucleation photon or 60 Gy or more of helium ion irradiation, virtually no cells are detected in the synthesis phase of the DNA cell cycle. It is unclear whether the absence of cycling cells after 20 Gy of photon irradiation is permanent or represents a transient cell cycle block, since these tumors were studied within 48 hours after irradiation. In contrast, all melanomas treated with helium ion had been irradiated several months prior to enucleation (mean, 2 years). In the latter group of tumors, the length of time between treatment and cell cycle analysis suggests that these cells had lost their reproductive integrity. These data were substantiated by tissue culture studies. Growth of tumor explants was significantly less ( $P < .007$ ) in irradiated than in nonirradiated melanomas.

The optimum technology used for measurement of cell cycle status

remains to be determined. Measurement of BrdUrd uptake using immunofluorescent microscopy on either standard sections or fine-needle biopsies can be performed. In general, flow cytometric analysis yields similar results. It is difficult with the latter technique to be certain that nontumor cells are not artifactuously counted in the cell cycle studies.

The incorporation of BrdUrd cell cycle analysis with fine-needle biopsy may be useful in the clinical management of irradiated melanomas that have questionable growth after treatment. In a few cases studied, results appeared to correlate with tumor growth activity.

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